

Engine Subsystem EDP doc

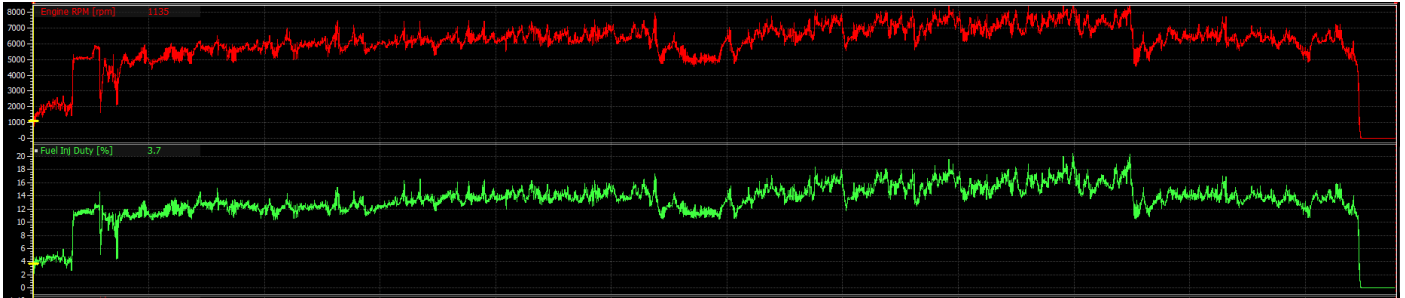
Fuel-efficiency calculation:

METHOD1:

Done using Fuel injector duty cycle-

Formula used for fuel consumed at a particular RPM:

$$\text{Fuel consumed (in cc)} = \frac{\text{RPM}}{60} \times \frac{\%dc}{100} \times 0.01 \times \frac{f}{60} \times \Delta t \times \frac{n}{2}$$



- RPM and Duty cycle data obtained from MoTec log files during test run
- A MATLAB code was written to calculate the fuel consumed during 30.637 minutes endurance run , extrapolating fuel consumption obtained from 10 minutes MoTec log data
- FOS of 1.2 on the Endurance lap time was considered for calculation

```
T= readtable("ALap.xlsx");
RPM=T.EngineRPM;
duty=T.FuelInjDuty;
time=T.Time;
num_cylinders = 4;
% fuel injector rate in cc/min
vol_rate=235;
minRPM0 = 17;
maxRPM0 = 1000;
minRPM1 = 1000;
maxRPM1 = 2000;
minRPM2 = 2000;
maxRPM2 = 3000;
minRPM3 = 4000;
maxRPM3 = 5000;
minRPM4 = 5000;
maxRPM4 = 6000;
minRPM5 = 6000;
maxRPM5 = 7000;
minRPM6 = 7000;
maxRPM6 = 8000;
rpm0 = T(RPM >= minRPM0 & RPM <= maxRPM0, :);
rpm1 = T(RPM >= minRPM1 & RPM <= maxRPM1, :);
rpm2 = T(RPM >= minRPM2 & RPM <= maxRPM2, :);
rpm3 = T(RPM >= minRPM3 & RPM <= maxRPM3, :);
rpm4 = T(RPM >= minRPM4 & RPM <= maxRPM4, :);
rpm5 = T(RPM >= minRPM5 & RPM <= maxRPM5, :);
rpm6 = T(RPM >= minRPM6 & RPM <= maxRPM6, :);

t0=height(rpm0)*0.002;
avg0=854;
fdc0=2.9;
%in fuel consumed in cc
fc0=(avg0/60)*2.9*0.0001*(num_cylinders/2)*(vol_rate/60)*t0

t1=height(rpm1)*0.002;
avg1=mean(rpm1.EngineRPM)
fdc1=3.9;
fc1=(avg1/60)*fdc1*0.0001*(num_cylinders/2)*(vol_rate/60)*t1

t2=height(rpm2)*0.002;
avg2=mean(rpm2.EngineRPM)
fdc2=4.8;
fc2=(avg2/60)*fdc2*0.0001*(num_cylinders/2)*(vol_rate/60)*t2

t3=height(rpm3)*0.002;
avg3=mean(rpm3.EngineRPM)
fdc3=11.3;
fc3=(avg3/60)*fdc3*0.0001*(num_cylinders/2)*(vol_rate/60)*t3
```

The code checks the amount of time the car has run in between each 1000 RPM starting from 0-1000RPM, 1000-2000RPM and so on.

Then an average RPM is calculated and the fuel consumed during the time of run in that mean RPM is obtained by mapping fuel injection duty cycle data

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```
t4=height(rpm4)*0.002;
avg4=mean(rpm4.EngineRPM)
fdc4=12.8;
fc4=(avg4/60)*fdc4*0.0001*(num_cylinders/2)*(vol_rate/60)*t4

t5=height(rpm5)*0.002;
avg5=mean(rpm5.EngineRPM);
fdc5=14.1;
fc5=(avg5/60)*fdc5*0.0001*(num_cylinders/2)*(vol_rate/60)*t5

t6=height(rpm6)*0.002;
avg6=mean(rpm6.EngineRPM)
fdc6=16.3;
fc6=(avg6/60)*fdc6*0.0001*(num_cylinders/2)*(vol_rate/60)*t6

total_time=t0+t1+t2+t3+t4+t5+t6
fc_total=fc1+fc2+fc3+fc4+fc5+fc6+fc0

x=[t0 t0+t1 t0+t1+t2 t0+t1+t2+t3 t0+t1+t2+t3+t4 t0+t1+t2+t3+t4+t5 t0+t1+t2+t3+t4+t5+t6]
y=[fc0 fc0+fc1 fc1+fc0+fc2 fc0+fc1+fc2+fc3 fc0+fc1+fc2+fc3+fc4 fc0+fc1+fc2+fc3+fc4+fc5 fc0+fc1+fc2+fc3+fc4+fc5+fc6]
figure;
plot(x, y, 'o'); % 'o' plots points as circles
xlabel('time(s)');
ylabel('Fuel_consumed(cc)');
title('Fuel_consumed vs time');
grid on;
hold on;

p = polyfit(x, y, 2); % 2nd-order polynomial fit
x_fit = linspace(min(x), max(x), 100); % Smooth range for plotting
y_fit = polyval(p, x_fit);

plot(x_fit, y_fit, '-r', 'LineWidth', 2);
legend('Data', 'Quadratic Fit');

t_new = 2205; %30.637 minutes endurance laptime from track data with 1.2 FOS gives 42.96138min=2568.972sec
fc_extrapolated = polyval(p, t_new);

fprintf('Extrapolated Fuel Consumed at t = %.2f is %.4f\n', t_new, fc_extrapolated);
```

```
x =
    0.7180    10.1060    19.0680    52.7080    202.9380    464.4980    566.3620
|
y =
    0.0232    0.8316    2.0982    25.8041    166.0893    475.8517    636.2506

Extrapolated Fuel Consumed at t = 2205.86 is 5519.8600
```

The fuel consumed value is 5519.86 cc or ml

METHOD 2:

Done using Fuel injector duty cycle-

Formula used for fuel consumed at a particular RPM:

$$\text{Fuel consumed (in cc)} = \frac{\text{RPM}}{60} \times \frac{\%dc}{100} \times 0.01 \times \frac{f}{60} \times \Delta t \times \frac{n}{2}$$

%dc= duty cycle percent

F flow rate of injector=235cc/min

0.01=10 millisecond (reference duty cycle for each injector)

n= number of engine cylinder, n/2 is considered as during each

Data Used- Kari motor Speedway, track data of car obtained by simulation in OptimumLap

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- Duty cycle from the duty cycle map of the car at various RPM was mapped with the RPMs at which the car ran during a single lap

A	B	C	D	E	F	G	H	I	J
Length	Time	Speed (angular)	Fuel injector duty(with compensation)	RPM	Fuel consumed (in ml)		Total length of 1 lap(m)		
0.18	0	6012.341741		12.9	6012			1032.53	
0.36	0.026535422	6155.746339		13.8	6156	0.029430595			
0.54	0.052452673	6295.718175		12.7	6296	0.027055324	Total fuel consumed (in ml)	255.0479	
0.72	0.077793709	6432.48819		12.8	6432	0.027238033	Endurance distance	22km	
0.9	0.102595935	6566.262315		13.1	6566	0.027852115			
1.08	0.126892866	6697.225119		14.4	6697	0.030590711	Number of laps to be taken=	21.30689	
1.26	0.150714676	6825.542795		13.7	6826	0.029084189			
1.44	0.174088645	6951.365638		14.7	6951	0.031181172	Fuel Efficiency of car(in km/lit)	4.048376	
1.62	0.197039534	7074.830099		14.5	7075	0.030738971	(in lit/km)	0.247013	
1.8	0.219589902	7196.060511		15.6	7196	0.033049489			
1.98	0.241760369	7315.170554		14.9	7315	0.031547925	Fuel required for endurance (in Lit)	5.434277	
2.16	0.263569844	7432.264489		15.1	7432	0.031953851			

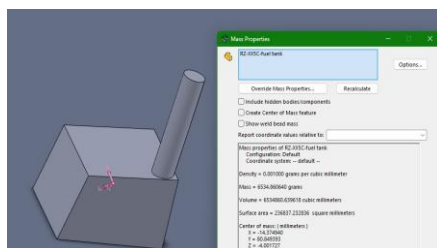
The fuel consumed value is 5434.27 cc or ml

Error in the fuel consumption between two methods is =(5519.86-5434.27)=85.59 cc

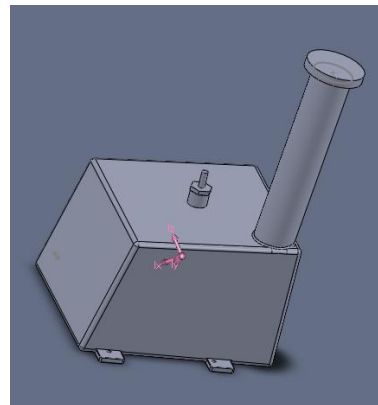
%Error=(85.59/5519.86)*100=1.55%

Fuel tank design

Old fuel tank

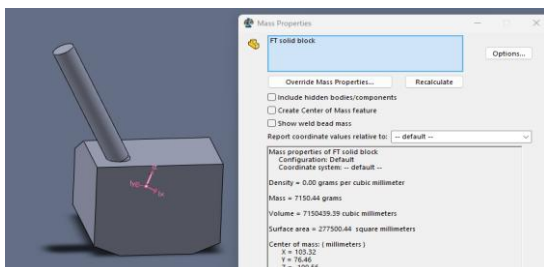


Old fuel tank volume model

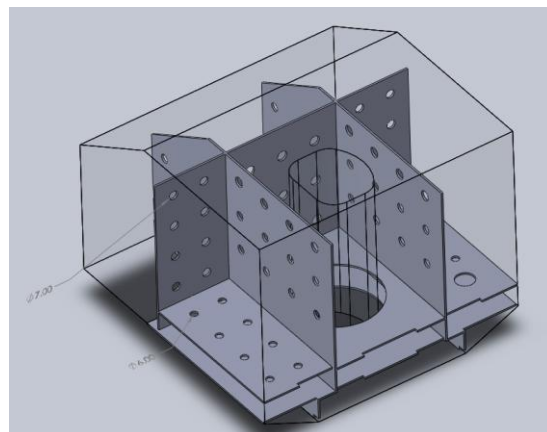


- Volume(without fuel pump) = 6.534lit
- Approximate Volume occupied by Fuel pump =250ml
- Actual Volume= 6.534-0.35=6.184lit
- The old tank did not have of any baffles

New fuel tank

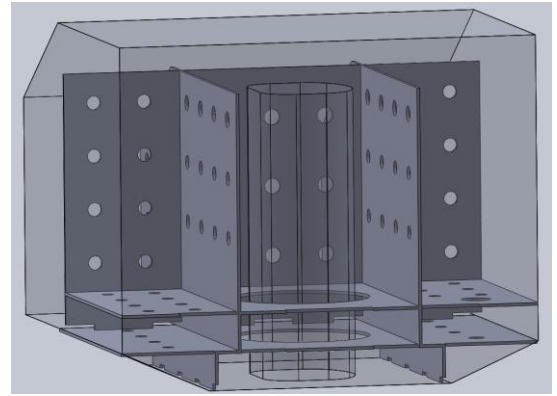
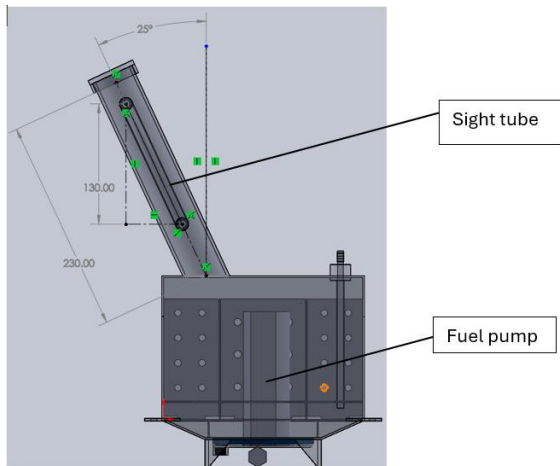


New fuel tank volume model



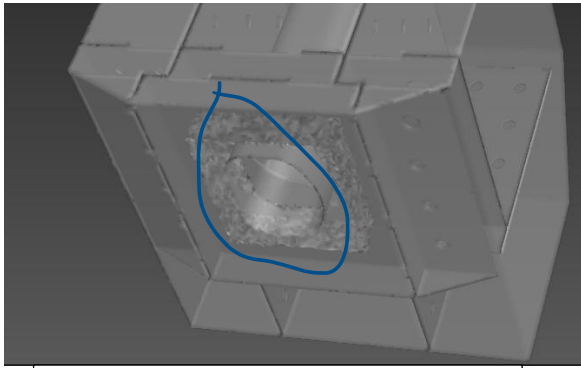
- Volume(without fuel pump)=7.15 lit, salvaged pump volume=0.250lit
- Volume occupied by baffles= 0.13 lit (1mm thick aluminum sheets used)
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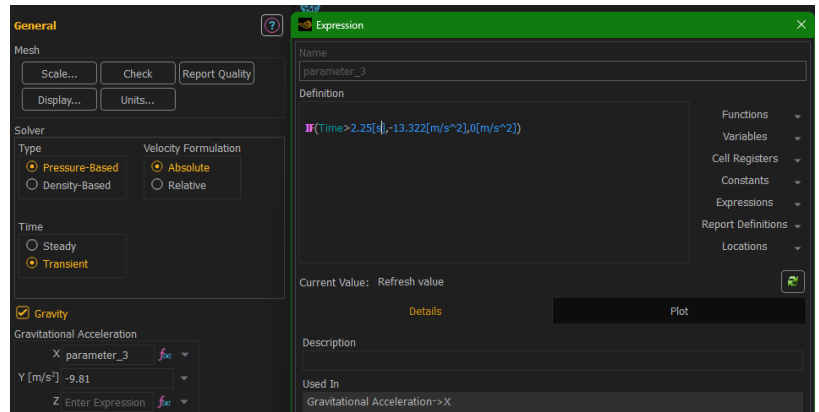


- Actual volume= $7.15-(0.25+0.13)=6.77\text{lit}$
- The bottom of the tank is made as a separate reservoir of volume close to 450ml, to prevent fuel starvation caused due to sloshing
- Bottom baffle plate hole diameter=6mm to prevent excess fuel leak from the bottom reservoir of tank , other baffle plate hole diameter=7mm
- The return line tube dumps the fuel directly to the bottom of the tank

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Region where fuel is monitored

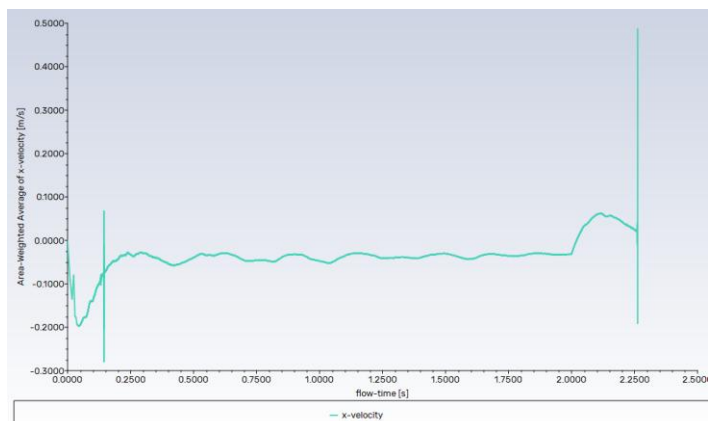
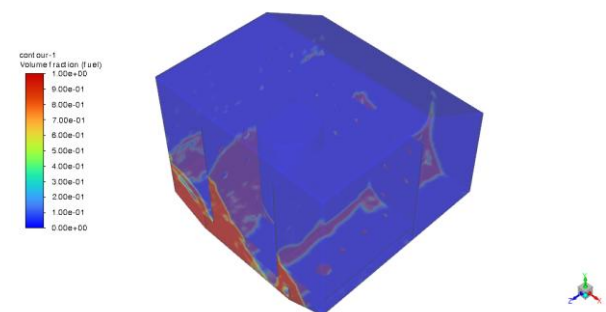
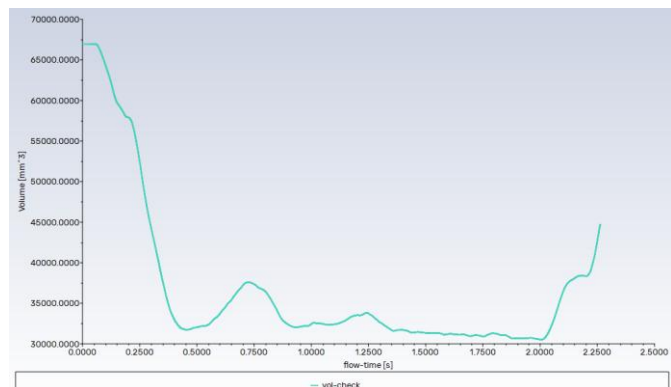
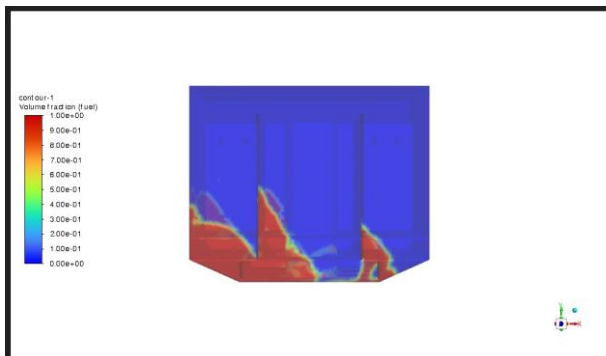


Sloshing analysis was done for a lateral acceleration of 13.322 m/s^2 (1.358g, obtained from the suspension team goal), by providing acceleration in the negative X direction according to geometry orientation

Results

Fuel volume near base region of the tank was observed, to check for fuel starvation, and the volume of fuel was observed to be stable at close to 31ml which is very close to the fuel pump requirement for such a small duration of time. **(Note: 30.555ml/s[110lit/hr] value derived from the fuel flow rate of GSX-r600 fuel pump)**

The X-direction velocity remains stable.



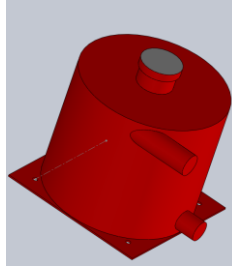
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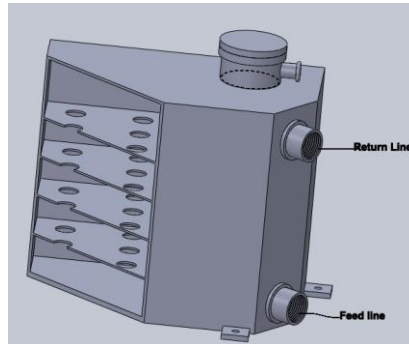
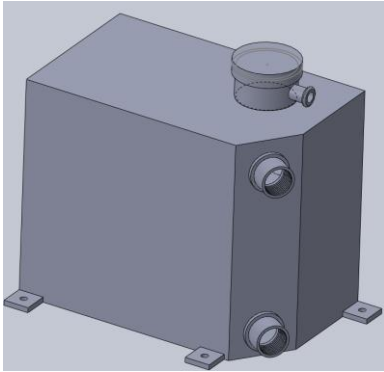
OIL TANK DESIGN

Old Tank

- Volume = 3.94lit
- Cylinder diameter = 180mm



New Oil Tank



Volume of tank was decided based on

- Engine oil requirement when the engine was a dry sump (from GSX R600 service manual)
- The volume of oil that will be in return and feed line

each of 1m long and internal diameter of 11mm,
therefore volume of oil in the lines = $2 \times \frac{\pi \times (0.011)^2}{4} =$
0.1906 liters

Necessary amount of engine oil

Oil change: 2 200 ml (2.3/1.9 US/Imp qt)

Oil and filter change: 2 500 ml (2.6/2.2 US/Imp qt)

Engine overhaul: 2 900 ml (3.1/2.6 US/Imp qt)

- When converting from wet to dry sump the oil volume is usually increased by 0.6-0.9 liters
Considering that the oil tank was designed with the volume= 3.636 liters
This includes volume occupied by baffles and the volume of oil that remains in the feed line =

$$0.127 + 0.1906 = 0.3176 \text{ liters}$$

Actual volume of oil available = $3.636 - 0.3176 = 3.3184$ liters

Which is 0.8184 liters more than the wet sump oil requirement of 2.5 liters

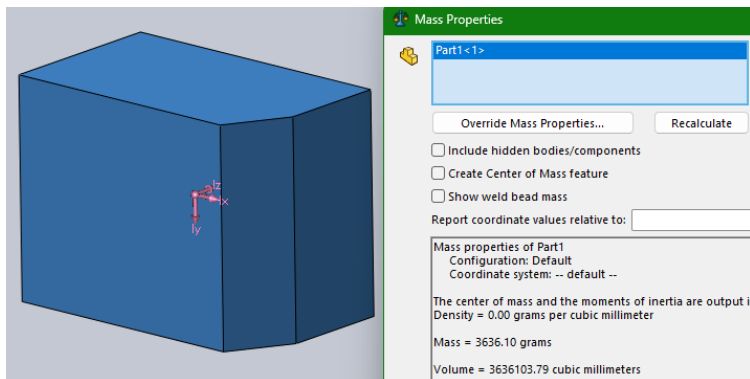
This volume is 0.304 liters lesser than the old tank

Design Features

- Horizontal baffles of hole diameter 15mm used,

25mm spacing between each baffle plate

- Baffle plates are placed right below opening of return line to minimize aeration
- New tank has width of 120mm for better packaging
- A flat side where the return and feed line adapters will be welded
- A positive stop opening near the cap is provided for catch can connection



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